The warning existed. The decision did not.



8 Jul 2022

Amarnath cave

Flash flood. No rain data from the upper reaches.



14 Aug 2025

Chasoti, Machail yatra

IMD alerts for the region; the yatra went on. About 61–64 dead.



26 Aug 2025

Adhkuwari, Vaishno Devi

Landslide at 14:40; at least 34 dead. Board: old track closed by ~12:00.

Each time, a warning reached some official. Nobody turned it into “close this stretch, move these people, now” — and nobody can show who decided.

Sources: Onmanorama, MoES (2022); ETV Bharat (2025); The Tribune (2025). Links in docs/research.md.

Who is affected, and where the gap is

IMD / GSI forecasts

District scale

RFID / CCTV crowd

No hazard input

SACHET mass alerts

Areas, not people in transit

Shrine-board control room

No record of who approved what

District alert

GAP: nobody owns “close this 500 m now”

Pilgrim on segment X

Who it protects: pilgrims on mountain tracks (Vaishno Devi cap: 50,000 a day), pony and palki operators, shopkeepers on the track, SDRF and police on duty.

Sources: The Tribune (SMVDSB RFID, 700+ CCTV, 50,000/day cap); ETV Bharat (SACHET); Business Standard, MoES (GSI bulletins). Gap: team inference from these. Links in docs/research.md.

YatraKavach: risk per 500 m, decided by a named officer



Where

35 segments on the three Vaishno Devi tracks. Hazard every hour from published Himalayan rain thresholds.

Who is there

SIMULATED

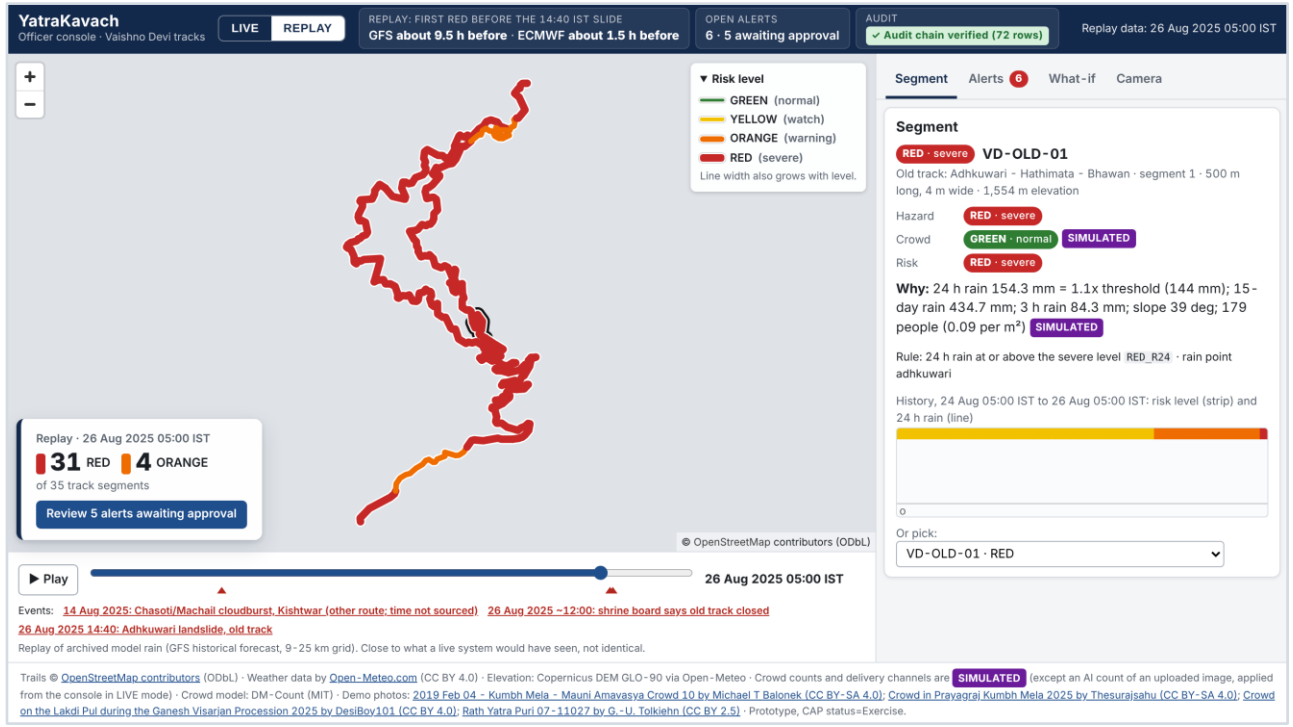
People per segment and the queue a hold creates. AI crowd count from a camera frame (DM-Count).

Who decides

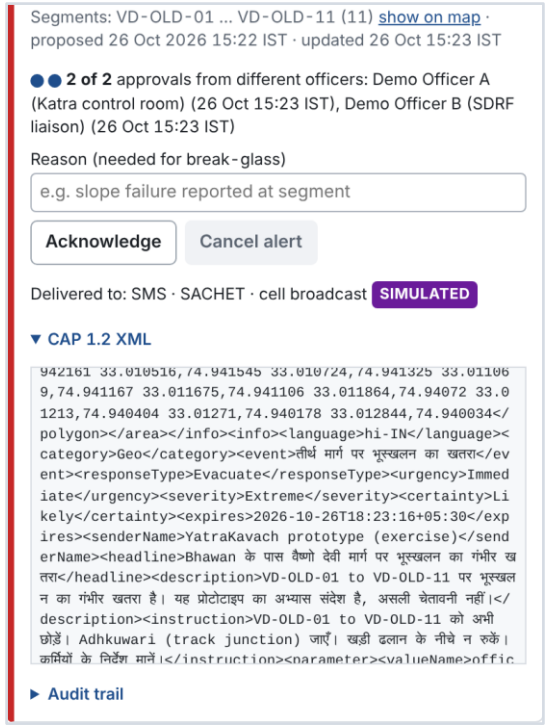
RED needs two officers; escalation after 10 min; break-glass with a reason. Hash-chained audit log.

CAP 1.2, the format NDMA SACHET uses · FastAPI · PostgreSQL / PostGIS · MapLibre · open source (MIT). Crowd counts on tracks and delivery channels are SIMULATED in the prototype; camera demo uses licensed festival photos.

Replay: 26 August 2025



26 Aug 05:00 IST: old track at Adhkuwari RED, with the reasons in numbers



RED alert: 2 of 2 officers, CAP in Hindi

9.5 h

RED before the slide (GFS)

1.5 h

with ECMWF or ERA5

674 → 0

on the slide segment if held at first ORANGE

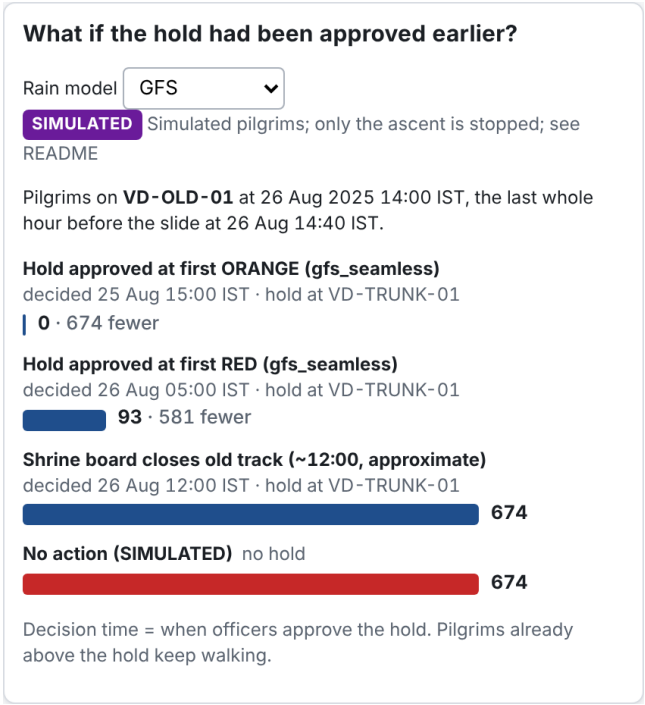
SIMULATED

Holding the ascent fills the Katra gate to 5.4 persons/m² inside a RED segment (SIMULATED): **closing a track moves the risk to the people below it.**

Replay of archived model rain (Open-Meteo historical forecast, CC BY 4.0; 9–25 km grid): close to what a live system would have seen, not identical. Screenshots: real API replay; approval timestamps are from the demo run.

Impact, KPIs and scale

KPI	Prototype result	Pilot target
Lead time of RED before a slide	1.5–9.5 h (26 Aug 2025 replay, 3 models)	≥ 3 h with radar input
Alerts with a named approver and time	100%	100%
Open alerts at peak (26 Aug, 14:00)	6: one per track and level	≤ 1 per track and level
Pilgrims on slide segment at 14:00 (SIMULATED)	674 no action → 0 if held at first ORANGE; 93 at first RED (GFS)	Decision within 1 h of ORANGE
Camera crowd count error	MAE 60 people/image (ShanghaiTech A)	Validated on SMVDSB CCTV
Time from proposal to decision	Logged per alert	Median < 10 min



Console what-if panel, GFS (SIMULATED pilgrims)

Scale: same engine for Amarnath, Machail, Char Dham, Sabarimala — only segments and thresholds change. **Sustainable:** open source, one small server, fits shrine-board control rooms and SDMA. Add Dogri, Kashmiri, Pahari with native speakers.

Implementation roadmap and what it needs

1 Pilot (one monsoon)

Old track + Himkoti with the SMVDSB control room and SDRF. Read-only, no public alerts; compare proposals with real decisions.

2 Calibrate

Local slide records, IMD radar feed, 3–5 rain gauges on the track, GSI susceptibility layer.

3 Integrate

RFID gate counts, CCTV density, approved CAP to SACHET, app and PA.

What it needs

Data-sharing MoU · one server · a field-officer champion · native-speaker review

Limits, shown openly

Thresholds not calibrated for these tracks · 9–25 km rain grid · crowd and delivery channels SIMULATED

Team and AI-use disclosure

RB

Ritik Bhardwaj

B.Tech IT, NIT Srinagar · solo entry, J&K

Problem framing, design decisions, validation, pitch.

Code: github.com/RitikBhardwaj-programmer/YatraKavach (MIT)

AI-use disclosure

Tool: Claude Code (Anthropic), as coding and research agent.

AI did: research report, most code, tests and docs, an independent code review.

Team did: chose the problem, set requirements, approved every design and fix, validated results.

Share (team estimate): about 60–65% of code and documents AI-generated under the team's direction.

Data and credits: Tracks © OpenStreetMap contributors (ODbL). Elevation: Copernicus DEM GLO-90 via Open-Meteo. Rain: Open-Meteo.com (CC BY 4.0), ECMWF IFS, NOAA GFS, ERA5. Thresholds: Shah et al. 2024; Dahal & Hasegawa 2008. Crowd model: DM-Count (MIT). Demo photos: Wikimedia Commons (CC BY 4.0 / CC BY 2.5 / CC BY-SA 4.0), authors in the repo. Pilgrim counts, holding queues and delivery channels are SIMULATED; all CAP messages are status=Exercise.